Theory

ISPhO

A3-1

Korean (Korea (Republic of))

링 공진기를 이용한 주파수 측정

Part A. 광섬유에서의 장 분포 (2.5점)

A1 (0.3pt)

$C_1 =$ $C_2 =$

A2 (0.5pt)

$k_{\text{in}} =$ $k_{\text{out}} =$

A3 (0.3pt)

A4 (0.3pt)

$B =$

A5 (0.3pt)

A6 (0.3pt)

A7 (0.3pt)

식:

A8 (0.2pt)

식:

Part B. 광섬유 공진기 투과율 (3.0점)

B1 (0.2pt)

식:

B2 (0.3pt)

$B_{\text{in}}(t) =$

B3 (0.3pt)

$B_{\text{in}}(t) =$

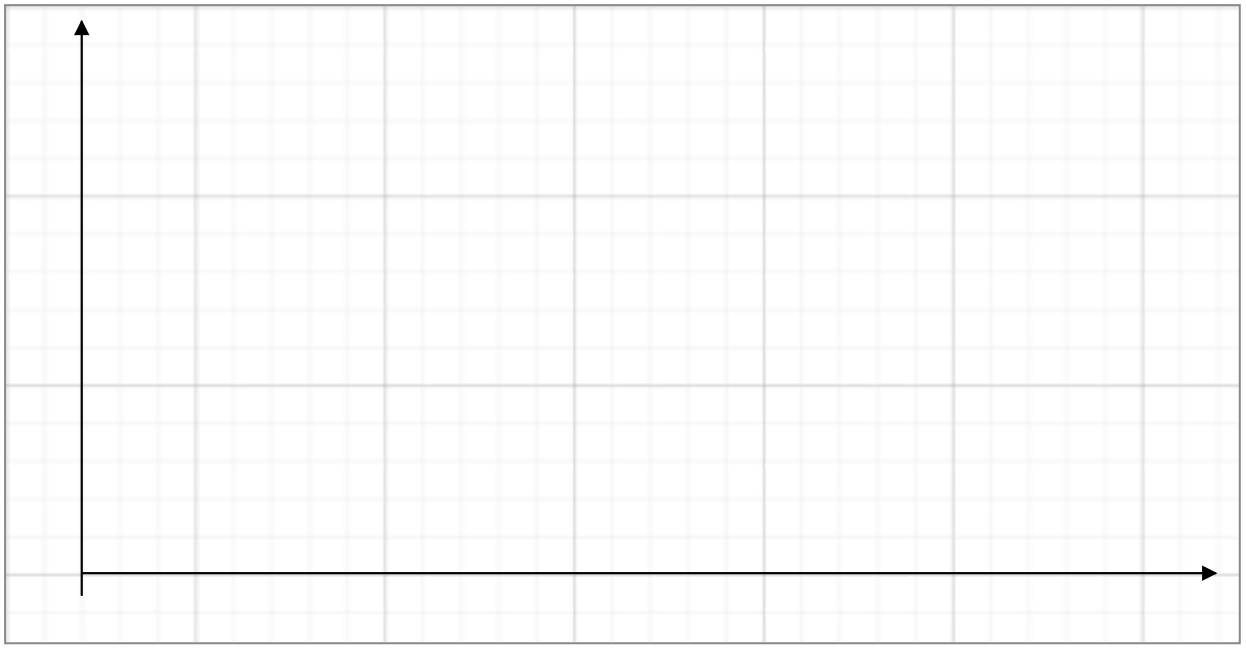
B4 (0.5pt)

$B_{\text{in}}(t) =$

B5 (0.5pt)

$N_2 =$

B6 (0.6pt)
그래프:

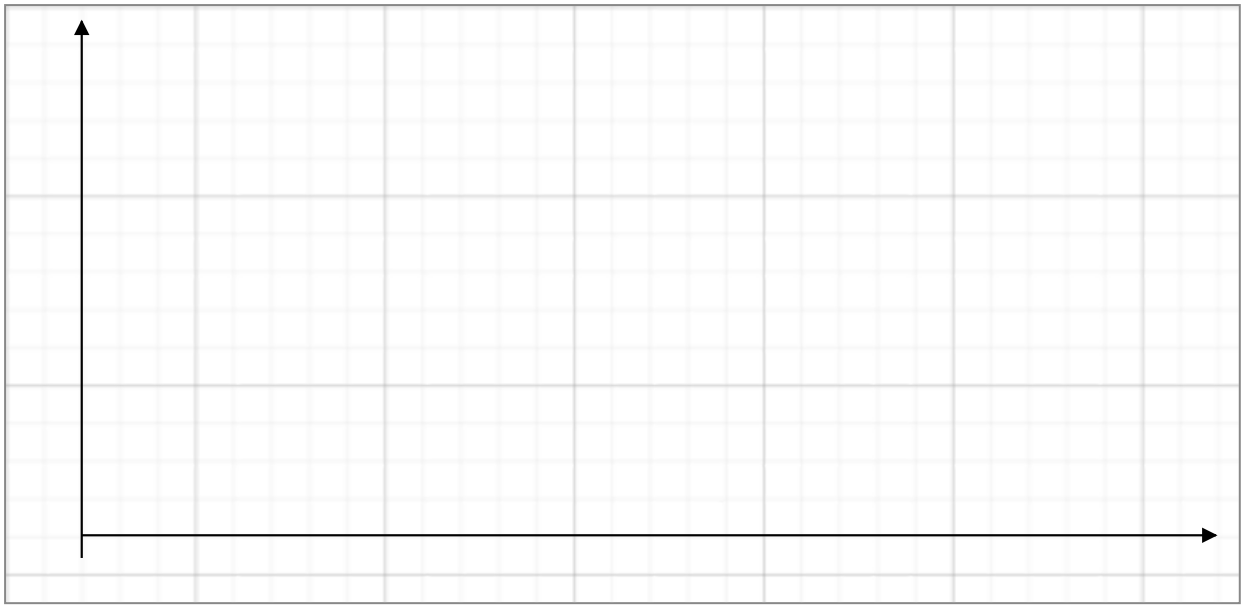


B7 (0.6pt)

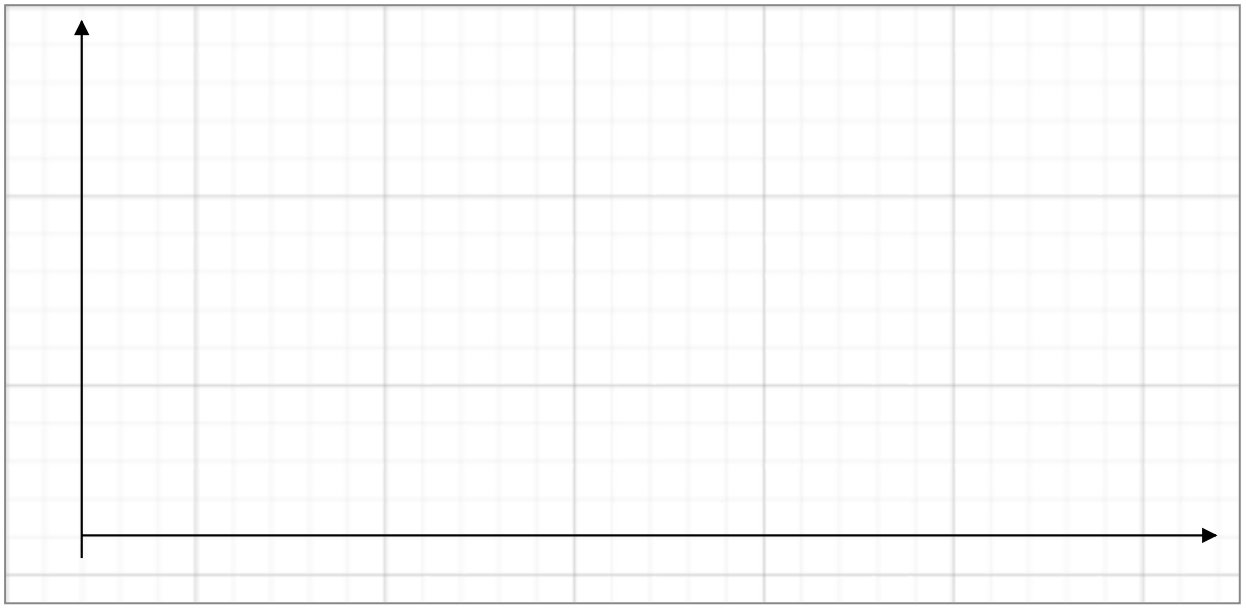
$Q =$

Part C. 마흐-젠더 간섭계의 보정 (2.0점)

C1 (0.5pt)
그래프:



C2 (1.0pt)
그래프:



C3 (0.5pt)

$\Delta\omega_{\mathrm{MZI}}/\omega_{\mathrm{MZI}} =$

Part D. 3차 비선형성과 솔리톤 (2.5점)

D1 (0.1pt)

$v_g =$

D2 (0.6pt)

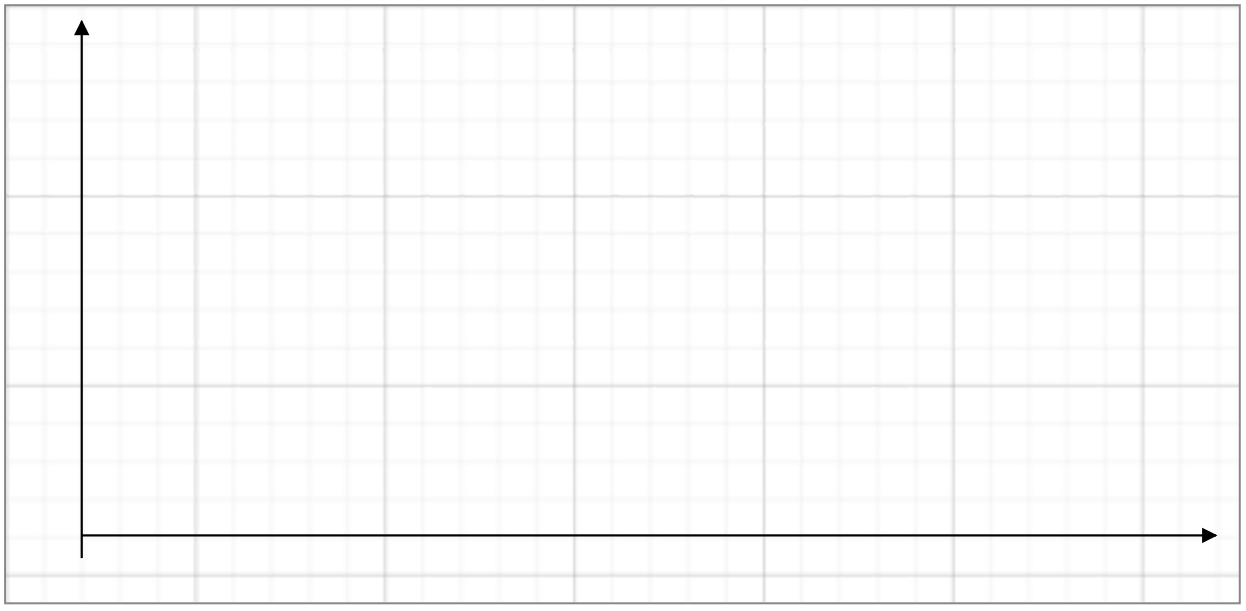
D3 (0.5pt)

$D_1 =$ $D_2 =$

D4 (0.3pt)

D_2에 대한 조건:

D5 (0.6pt)
그래프:



D6 (0.2pt)

$\Delta\omega =$

D7 (0.2pt)

$\tau_s =$